



Green's function and representation theorem

Chapter 3

Green's function is useful when considering the seismic wavefield excited by an event. This chapter explains a framework for interpreting seismic wavefields excited by various events, such as earthquakes and volcanic eruptions, using the representation theorem, which can be regarded as a natural extension of Huygens's principle. First, an acoustic (scalar) wave treatment is explained; then, an elastic (vector) wave treatment is explained briefly.

§3.1 A solution of the wave equation in 1-D medium

First, let us consider the simplest case of a wave equation: the 1-D wave equation. Here, we consider the propagation of acoustic waves. The elastic constant κ and density ρ are homogeneous for simplicity. The pressure fluctuation p satisfies the following wave equation:

$$\frac{1}{\alpha^2} \frac{\partial^2 p(x, t)}{\partial t^2} - \frac{\partial^2 p}{\partial x^2} = 0, \quad (3.1)$$

where α is the sound speed given by $\alpha = \sqrt{\kappa/\rho}$.

To understand, the wave equation is mapped into the time-frequency domain by the Fourier transform. The Fourier component $P(k, \omega)$ of the pressure fluctuation becomes a function of wavenumber k and the angular frequency ω as:

$$\left(\frac{\omega^2}{\alpha^2} - k^2 \right) P(k, \omega) = 0. \quad (3.2)$$

Thus P must satisfy a dispersion relation $\omega^2/\alpha^2 - k^2 = 0$. Here we consider one Fourier component $P(k, \omega)e^{i(\omega t + kx)}$. Because the dispersion relation leads to $k = \pm\omega/\alpha$, the Fourier component

can be rewritten as $P(k, \omega)e^{i(\omega(t \pm x/\alpha))}$, which represents the propagation toward the positive and negative directions of the x axis, respectively.

d'Alembert solution

Let us evaluate the behavior of the solution more mathematically. With changes of variables, $\xi = x - \alpha t$, $\eta = x + \alpha t$, the solution of the 1-D wave equation can be represented by the arbitrary function ϕ and ψ as,

$$p(x, t) = \phi(x - \alpha t) + \psi(x + \alpha t). \quad (3.3)$$

The first term of the right-hand side represents the propagation toward the positive direction along the x axis, whereas the second term represents the propagation toward the negative direction.

Initial value problem

Let us consider how to solve the problem for the initial value at $t = 0$ given by

$$p(x, 0) = p_0(x), \quad (3.4)$$

$$\left. \frac{\partial p}{\partial t} \right|_{t=0} = q_0(x). \quad (3.5)$$

By comparing the initial value with the d'Alembert solution as

$$p_0 = \phi(x, 0) + \psi(x, 0) \quad (3.6)$$

$$q_0 = -\alpha \left(\frac{\partial \phi}{\partial x} - \frac{\partial \psi}{\partial x} \right). \quad (3.7)$$

Integrating equation 3.7 leads to the solutions for ϕ and ψ . By the insertion of ϕ and ψ , we obtain the solution $p(x, t)$ as,

$$p(x, t) = \frac{1}{2} [p_0(x - \alpha t) + p_0(x + \alpha t)] + \frac{1}{2\alpha} \int_{x-\alpha t}^{x+\alpha t} q_0(x') dx'. \quad (3.8)$$

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Problem 3.1

1. Derive equation 3.3.
2. When p meets the initial condition at $t = 0$ given by

$$p(x, 0) = e^{-\frac{x^2}{\sigma^2}} \quad (3.9)$$

$$\left. \frac{\partial p(x, 0)}{\partial t} \right|_{t=0} = 0 \quad (3.10)$$

solve and plot the solution.

3. When p meets the initial condition at $t = 0$ given by

$$p(x, 0) = 0, \quad (3.11)$$

$$\left. \frac{\partial p(x, t)}{\partial t} \right|_{t=0} = e^{-\frac{x^2}{\sigma^2}}, \quad (3.12)$$

solve and plot the solution.

§3.2 Acoustic Green's function

An external force can represent the excited waves in the context of seismic waves caused by geophysical phenomena such as earthquakes and volcanoes. In this case, we can evaluate the excited wave motion by considering the impulse response (Green's function) to the external force and convolving it with the spatiotemporal distribution of the external force. This section first shows the features of Green's function.

First, let us consider acoustic wave propagation for essential understanding. The equations of motion and Hook's law ($\kappa \nabla \cdot \mathbf{s} = -p$) lead to a wave equation concerning pressure perturbation $p(\mathbf{x}, t)$ as follows:

$$-\nabla \cdot \frac{\nabla p(\mathbf{x}, t)}{\rho_0(\mathbf{x})} + \frac{1}{\kappa(\mathbf{x})} \frac{\partial^2 p(\mathbf{x}, t)}{\partial t^2} = -\nabla \cdot \left(\frac{\mathbf{f}(\mathbf{x}, t)}{\rho_0(\mathbf{x})} \right). \quad (3.13)$$

Here we consider a Green's function $g(\mathbf{x}, t; \xi, \tau)$, which is an impulse response for an impulsive force $\delta(\mathbf{x} - \xi)\delta(t - \tau)$ ^{note 1)} as

$$-\nabla \cdot \frac{\nabla g(\mathbf{x}, t; \xi, \tau)}{\rho_0(\mathbf{x})} + \frac{1}{\kappa(\mathbf{x})} \frac{\partial^2 g(\mathbf{x}, t; \xi, \tau)}{\partial t^2} = -\delta(\mathbf{x} - \xi)\delta(t - \tau). \quad (3.14)$$

Problem 3.2

Explain the physical meaning of the external force term $-\delta(\mathbf{x} - \xi)$ of equation 3.14 (divergence of the particle velocity $\nabla \cdot \mathbf{v}$).

If the boundary condition is time-independent, the Green's function exhibits time invariance as,

$$g(\mathbf{x}, t; \xi, \tau) = g(\mathbf{x}, t - \tau; \xi, 0). \quad (3.15)$$

Therefore, the time difference $t - \tau$ is sufficient to represent this problem.

note 1)

$$f(0) = \int f(\mathbf{x})\delta(\mathbf{x})dV.$$

Therefore, note that the dimension of the delta function in 3D is $1/\text{m}^3$.

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The pressure field $p(\mathbf{x}, t)$ can be represented by the superposition of Green's function ($g(\mathbf{x}, t - \tau; \xi, 0)$).

$$p(\mathbf{x}, t) = \int_V g(\mathbf{x}, t; \xi, \tau) \nabla_{\xi} \cdot \left(\frac{\mathbf{f}}{\rho_0} \right) dV(\xi) d\tau. \quad (3.16)$$

Next, the representation of the equation in the frequency domain is considered. $P(\omega, \mathbf{x})$ shows the Fourier transform of pressure p , and ω is the angular frequency^{note 2)}

We do not consider hydrostatic pressure here.

$$\nabla \cdot \frac{\nabla P(\mathbf{x}, \omega)}{\rho_0(\mathbf{x})} + \frac{\omega^2}{\kappa(\mathbf{x})} P(\mathbf{x}, \omega) = F(\mathbf{x}, \omega), \quad (3.17)$$

where F is the Fourier transform of the external force term $\nabla \cdot \left(\frac{\mathbf{f}(\mathbf{x}, t)}{\rho_0(\mathbf{x})} \right)$.

Green's function in the frequency domain satisfies the following equation,

$$\nabla \cdot \frac{\nabla G(\mathbf{x}, \xi, \omega)}{\rho_0(\mathbf{x})} + \frac{\omega^2}{\kappa(\mathbf{x})} G(\mathbf{x}, \xi, \omega) = \delta(\mathbf{x} - \xi), \quad (3.18)$$

where $G(\mathbf{x}, \xi, \omega)$ is Green's function in the frequency domain. Then, the pressure field P excited by an arbitrary force distribution F can be represented by a convolution between Green's functions and the force distribution as,

$$P = \int_V G(\mathbf{x}, \xi, \omega) F(\xi, \omega) dV(\xi). \quad (3.19)$$

§3.3 Green's function in an infinite homogeneous medium

To understand the behaviors of Green's function, this section explains the explicit representation of Green's function. First, let us summarize the basics of the wave equation.

^{note 2)}Take care regarding the definitions of the sign and the normalization of the Fourier transform because the definition depends on the field. The Fourier transform $\mathcal{F}$ and inverse Fourier transform $\mathcal{F}^{-1}$ are defined as

$$U(\omega) = \mathcal{F}(u) \equiv \int_{-\infty}^{\infty} u(t) e^{-i\omega t} dt,$$

$$u(t) = \mathcal{F}^{-1}(U) \equiv \frac{1}{2\pi} \int_{-\infty}^{\infty} U(\omega) e^{i\omega t} d\omega.$$

Details of the definition in this text are shown in the appendix 3.E.

3.3.1 Derivations of Green's function in a 1-D medium

For an infinite homogeneous medium, equation 3.18 can be simplified as,

$$\frac{1}{\rho_0} \frac{\partial^2 G^{1D}}{\partial z^2} + \frac{\omega^2}{\kappa} G^{1D} = \delta(z). \quad (3.20)$$

Problem 3.3

1. Solve the equation except for $z = 0$.
2. At $z = 0$, the equation is singular because of the delta function. Integrate equation 3.20 in a range of $-\epsilon/2 \leq z \leq \epsilon/2$. Then derive the following equation,

$$\left[\frac{\partial G^{1D}}{\partial z} \right]_{-\epsilon/2}^{+\epsilon/2} = \rho_0. \quad (3.21)$$

Note that integration of Green's function in the infinitesimal range is negligible from equation 3.19.

3. By continuation of the two solutions at $z = 0$, derive the following result of 1-D Green's function as,

$$G^{1D}(z, \omega) = \frac{\rho_0 i}{2k} e^{-ik|z|}. \quad (3.22)$$

4. Inverse Fourier transform of G^{1D} show

$$g^{1D}(z, t) = \begin{cases} 0 & t < |z|/\alpha \\ -\frac{\alpha}{2} \rho_0 & t \geq |z|/\alpha. \end{cases} \quad (3.23)$$

5. Problem 3.1 (3) gave the initial velocity at $t = 0$. Compare the solution to the above solution of this problem and interpret the physical relation.

Derivation of 1D Green's function using complex integration

Here, we calculate a Fourier transform in the spatial direction and use complex integration to find G^{1D} . For simplicity, we define k_0 as $k_0^2 \equiv \rho_0 \omega^2 / \kappa$.

Considering the Fourier transform of G^{1D} with respect to space (z), we denote its Fourier component as $\tilde{G}$. When we consider the Fourier transform of the wave equation, the spatial derivative becomes ik using the wavenumber k , thus:

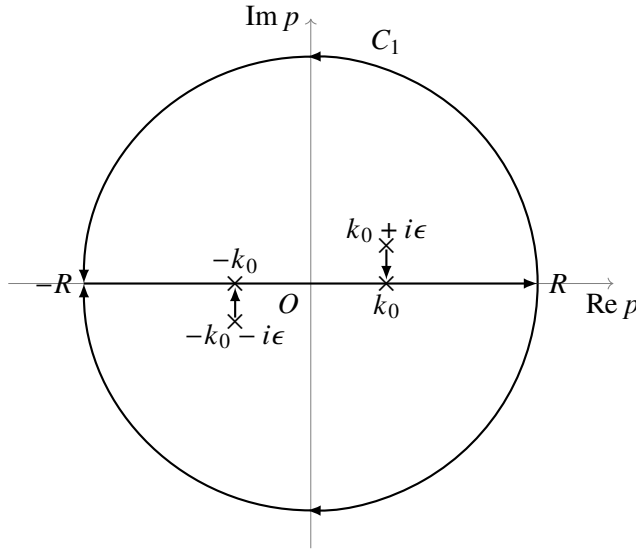
$$\tilde{G} = \frac{\rho_0}{k_0^2 - k^2} \quad (3.24)$$

By considering the inverse Fourier transform, we can find G^{1D} :

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$$G^{1D} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{\rho_0}{k_0^2 - k^2} e^{ikz} dk \quad (3.25)$$

We need to evaluate this integral using residues. However, as it stands, the first-order poles ($k = \pm k_0$) lie on the real axis, making evaluation difficult. Therefore, we consider causality and shift the poles to $\pm k_0 \pm i\epsilon$, taking the limit as ϵ approaches zero. Physically, this corresponds to a wave propagating from the origin to the future, with a slight attenuation in amplitude.



We can calculate G^{1D} as

$$G^{1D}(z, \omega) = \frac{\rho_0 i}{2k_0} e^{-ik_0|z|} \quad (3.26)$$

by evaluating the residue.

3.3.2 Derivations of Green's function in a 2-D medium

Here, we consider a cylindrical coordinate (r, ϕ) with the origin at $\mathbf{x}i$.

$$\frac{1}{\rho_0} \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial G^{2D}}{\partial r} \right) + \frac{\omega^2}{\kappa} G^{2D} = \delta(\mathbf{x} - \mathbf{x}_0). \quad (3.27)$$

1. Except for $r = 0$, a solution of equation 3.27 can be represented by the superposition of the 0th order Bessel function of the first kind $J_0(r)$ and the 0th order Neumann function $N_0(r)$.
2. At $r = 0$, the equation is singular. Then integrate equation 3.27 over an infinitesimal circular area (C) around the origin. Using Gauss's divergence theorem, show

$$\int_C \frac{\partial G^{2D}}{\partial r} dl = \rho_0. \quad (3.28)$$

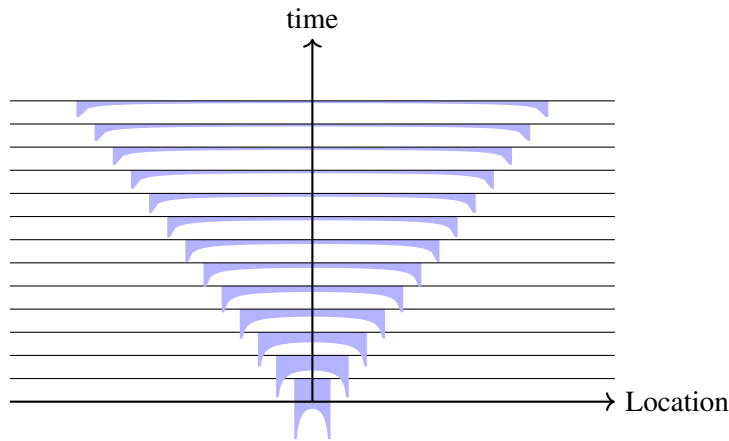


Figure 3.1: Propagation of 2-D Green's function.

3. By continuing the two solutions at the origin $r = 0$, derive the following result for the 2-D Green's function as

$$G^{2D} = -i \frac{\rho_0}{4} H_0^{(2)}(kr). \quad (3.29)$$

Here $H_0^{(2)}$ is Hankel's function of the second kind.

4. In the time domain, the corresponding Green's function can be written by,

$$g^{2D}(r, t) = -\frac{\rho_0}{2\pi} \frac{H(t - r/\alpha)}{\sqrt{t^2 - r^2/\alpha^2}}, \quad (3.30)$$

where H is the Heaviside step function.

Problem 3.4

1. Show that Hankel function of the first kind can be approximated by a cylindrical wave in far field.
2. Interpret distance dependence of the amplitude based on the conservation of energy.
3. Show the amplitude of 2D Green's function in time domain (equation 3.30) is proportional to $1/\sqrt{r}$ if the distance r is enough long.

3.3.3 Green's function of a 3-D medium

This subsection explains Green's function in a 3-D medium, which describes 3-D wave propagation for the forcing $\delta(\mathbf{x})\delta(t)$.

$$-\frac{1}{\rho_0} \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial g^{3D}}{\partial r} \right) + \frac{1}{\kappa} g^{3D} \frac{\partial^2}{\partial t^2} = -\delta(\mathbf{x})\delta(t) \quad (3.31)$$

To consider the amplitude, first, let us consider the d'Alembert solution^{note 3)}. With a change in the variable as $p = \bar{p}/r$, the solution is given by

$$p = \frac{p_0(t - r/\alpha)}{r} + \frac{p_1(t + r/\alpha)}{r}, \quad (3.32)$$

which represents spherical waves outward and inward. The amplitude decays with $1/r$. The distance-dependent is also given by energy conservation on the expanding (or shrinking) wavefront.

Green's function in a 3-D medium is given by

$$g^{3D}(r, t) = -\frac{\rho_0}{4\pi} \frac{\delta(t - r/\alpha)}{r}. \quad (3.33)$$

See Problem 3.5 for the derivation.

Problem 3.5

Here we consider a spherical coordinate (r, θ, ϕ) with origin at ξ .

$$\frac{1}{\rho_0} \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial G^{3D}}{\partial r} \right) + \frac{\omega^2}{\kappa} G^{3D} = \delta(\mathbf{x} - \xi) \quad (3.34)$$

1. With a change of variable as $G^{3D} = \bar{G}/r$, rewrite the above equation.
2. Solve the equation except $r = 0$.
3. At $r = 0$, the solution is singular. Integrate equation 3.18 within an infinitesimal sphere. Then show the following equation

$$\int_{\Sigma} \frac{\partial G^{3D}}{\partial r} d\Sigma = \rho_0, \quad (3.35)$$

using Gauss's divergence theorem.

4. By continuation of the two solutions at the origin $r = 0$ derive the following result of 3-D Green's function as,

$$G^{3D} = -\frac{\rho_0}{4\pi} \frac{e^{-ikr}}{r} \quad (3.36)$$

5. Calculate inverse Fourier transform the above equation.

^{note 3)}In general, the behaviors of solutions of wave equations in even dimensions are quite different from those in odd dimensions. The former is localized close to the wavefront and lasts for a long time.

3.3.4 [†]Green function in n -dimension space

The 2-D Green's function behaves quite differently from the 1-D and 3-D Green's functions. In this section, we will investigate these differences by examining the properties of the Green's function for the general n -dimensional wave equation. This section contains advanced material, so you may skip it.

Relationship between 3-D and 1-D Green's functions

In the 3-D case, we consider

$$\frac{1}{\rho_0} \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial G^{3D}}{\partial r} \right) + \frac{\omega^2}{\kappa} G^{3D} = \frac{\delta^+(r)}{4\pi r^2}. \quad (3.37)$$

By defining $G^{3D} = \Psi/r$ and rearranging the equation, we obtain

$$\begin{aligned} \frac{1}{\rho_0} \frac{\partial^2 \Psi}{\partial r^2} + \frac{\omega^2}{\kappa} \Psi &= \frac{\delta^+(r)}{4\pi r} \\ &= -\frac{1}{4\pi} \frac{d\delta^+(r)}{dr}. \end{aligned} \quad (3.38)$$

Looking closely at this equation, the left-hand side is in the form of a 1-D wave equation. Therefore, we can see that Ψ can be written in the form of a convolution with respect to r between the 1-D Green's function and the derivative of the delta function:

$$\begin{aligned} \Psi &= -\frac{1}{4\pi} G^{1D} * \frac{d\delta^+}{dr} \\ &= -\frac{2}{4\pi} \frac{\partial G^{1D}}{\partial r} \\ &= -\frac{1}{2\pi} \frac{\partial G^{1D}}{\partial r}. \end{aligned} \quad (3.39)$$

Here, we multiplied by 2 at the end because the corresponding Green's function G^{1D} is defined on both sides of 0 with respect to the variable. Ultimately, the 3-D and 1-D Green's functions can be connected by the relation

$$G^{3D} = -\frac{1}{2\pi r} \frac{\partial G^{1D}}{\partial r}. \quad (3.40)$$

This can be confirmed from the Green's function derived previously using a different method.

Relationship between n -D and $(n+2)$ -D Green's functions

For the n -D Green's function, we consider the n -D wave equation. Assuming the Green's function is symmetric about the origin, integrating over all directions other than r in polar coordinates yields

$$S_n \frac{1}{r^{n-1}} \frac{\partial}{\partial r} \left(r^{n-1} \frac{\partial G^n}{\partial r} \right) + k_0^2 G^n = \frac{\delta^+(r)}{\rho_0 r^{n-1}}. \quad (3.41)$$

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Here, S_n represents the surface area of the n -dimensional unit sphere, which appears because we fixed r and integrated over the other coordinates. Additionally, S_n satisfies the recurrence relation

$$S_n = \frac{2\pi^{n/2}r^{n-1}}{\Gamma(\frac{n}{2})}, \quad (3.42)$$

and it also satisfies the relationship

$$S_{n+2} = \frac{2\pi}{n}S_n. \quad (3.43)$$

The 3-D wave equation could be connected to the 1-D solution by a simple relation. Can we connect the $(n+2)$ -D solution and the n -D solution in general? In fact, it is possible. By a somewhat heuristic approach, if we define Ψ^n as

$$\Psi^n \equiv \frac{1}{r} \frac{\partial G^n}{\partial r} \quad (3.44)$$

and apply the $(n+2)$ -D Laplacian, simple partial differentiation leads to

$$\frac{1}{r^{n+1}} \frac{\partial}{\partial r} \left(r^{n+1} \frac{\partial \Psi^n}{\partial r} \right) = \frac{1}{r} \frac{\partial}{\partial r} \left(\frac{1}{r^{n-1}} \frac{\partial}{\partial r} \left(r^{n-1} \frac{\partial \Psi^n}{\partial r} \right) \right). \quad (3.45)$$

In other words, if we take the partial derivative of both sides of the n -D wave equation (eq. 3.41) with respect to r and divide by r , we can write

$$\frac{1}{r^{n+1}} \frac{\partial}{\partial r} \left(r^{n+1} \frac{\partial \Psi^n}{\partial r} \right) + k_0^2 \Psi^n = \frac{1}{\rho_0 S_n} \frac{1}{r} \frac{d}{dr} \left(\frac{\delta^+}{r^{n-1}} \right). \quad (3.46)$$

Let us evaluate the right-hand side. Omitting the constants, we obtain

$$\frac{1}{r} \frac{d}{dr} \left(\frac{\delta^+}{r^{n-1}} \right) = -(n-1) \frac{\delta^+}{r^{n+1}} + \frac{d\delta^+}{dr} \frac{1}{r^n} = -n \frac{\delta^+}{r^{n+1}}. \quad (3.47)$$

Here we used the derivative formula for the delta function. Rearranging the equation using $S_{n+2} = 2\pi S_n/n$, we obtain

$$\frac{1}{r^{n+1}} \frac{\partial}{\partial r} \left(r^{n+1} \frac{\partial \Psi^n}{\partial r} \right) + k_0^2 \Psi^n = -2\pi \frac{\delta^+}{\rho_0 S_{n+2} r^{n+1}}. \quad (3.48)$$

From this, we can see that

$$G^{n+2} = -\frac{1}{2\pi} \frac{1}{r} \frac{\partial G^n}{\partial r} \quad (3.49)$$

satisfies the $(n+2)$ -D wave equation. This means that for odd-dimensional Green's functions, the energy is concentrated at the wavefront, and Huygens' principle can be applied (secondary wave sources can be uniquely determined). In the case of odd dimensions, the Green's function can be written as

$$G^n = \left(-\frac{1}{2\pi} \frac{1}{r} \frac{\partial}{\partial r} \right)^{\frac{n-1}{2}} G^{1D}. \quad (3.50)$$

In the case of even dimensions, fractional derivatives formally appear, indicating that it cannot be simply connected to G^{1D} . In general, it can be written as

$$G^n = -\frac{i\rho_0}{4} \left(\frac{1}{2\pi} \frac{k}{r} \right)^{\frac{n}{2}-1} H_{\frac{n}{2}-1}^{(2)}(kr). \quad (3.51)$$

Here, $H_v^{(2)}$ is the Hankel function of the second kind^{note 4)}.

Problem 3.6

1. The pressure response for forcing $f(\mathbf{x}) = \delta(z)$ in a 3-D medium can be interpreted by 1-D Green's function. Based on this fact, show the relation between 1-D Green's function and 3-D Green's function.
2. Equation 3.30 shows 2-D Green's function does not go zero after the arrival of the wavefront. Derive 2-D Green's function from the 3-D Green's function based on 3-D Green's function with a line source given by $\delta(r)$.

^{note 4)}It can be derived using the equations in Appendix 3.C.

§3.4 Green's function in a homogeneous elastic medium

Let us consider the equation of motions

$$\rho \frac{\partial^2}{\partial t^2} s_i = \frac{\partial}{\partial x_j} T_{ij} + f_i. \quad (3.52)$$

Here we neglect self-gravitation and initial stress.

For an **isotropic and homogeneous elastic body**, the equation can be simplified as,

$$\begin{aligned} \rho \partial_t^2 s &= (\lambda + \mu) \nabla (\nabla \cdot s) + \mu \nabla^2 s + f \\ &= \underbrace{(\lambda + 2\mu) \nabla (\nabla \cdot s)}_{P\text{wave}} - \underbrace{\mu (\nabla \times \nabla \times s)}_{S\text{wave}} + f \end{aligned} \quad (3.53)$$

The first term of the right-hand side represents the P wave with volumetric changes, whereas the second one represents the S wave with shear deformation. Later we consider a homogeneous and isotropic medium

3.4.1 Elastic potential: Separation between P wave and S wave

In order to clarify the perspective, displacement s is written by scalar potential ϕ ^{note 5)} and vector potential ψ using Helmholtz's theorem as

$$s = \nabla \phi + \nabla \times \psi. \quad (3.54)$$

I note that vector potential ψ has ambiguity for the choice of the reference. Vector potential $\psi' \equiv \psi + \nabla \chi$ leads to $\nabla \times \psi' = \nabla \times \psi$. For static magnetic field, we choose a vector potential as $\nabla \cdot \psi$ in general. In the case of an electromagnetic field, vector potential is related to the scalar potential for each other, whereas, in the case of elastic deformation, they are independent, as shown in later sections. In the case of an elastic wave field, we can separate the P wave and S wave without the choice of a reference of vector potential as $\nabla \times \psi = 0$. For example, for a stratified medium described in later chapters, a form of vector potential as $\nabla \times (\psi + \nabla \times \chi)$ becomes convenient because the two terms (ϕ and χ) represent horizontally polarized S waves (SH) and vertically polarized S waves (SV), respectively.⁽⁴⁾

Insertion of equation 3.54 into equation 3.53 leads to

$$\begin{cases} \nabla^2 \left(\rho \frac{\partial^2 \phi}{\partial t^2} - (\lambda + 2\mu) \nabla^2 \phi \right) = \nabla \cdot f \\ \nabla \times \nabla \times \left(\rho \frac{\partial^2 \psi}{\partial t^2} + \mu \nabla \times \nabla \times \psi \right) = \nabla \times f \end{cases} \quad (3.55)$$

^{note 5)}Based on the definition, the scalar potential ϕ can be related to pressure $p = -\rho \alpha^2 \nabla^2 \phi$.

where α is P-wave velocity and β is S-wave velocity ($\lambda + 2\mu = \rho\alpha^2$, $\mu = \rho\beta^2$).

A formula of vector analysis of

$$\nabla \times \nabla \times \boldsymbol{\psi} = \nabla(\nabla \cdot \boldsymbol{\psi}) - \nabla^2 \boldsymbol{\psi} \quad (3.56)$$

and $\nabla \times [\nabla(\nabla \cdot \boldsymbol{\psi})] = 0$ lead to

$$\nabla \times \nabla \times \left(\rho \frac{\partial^2 \boldsymbol{\psi}}{\partial t^2} - \mu \nabla^2 \boldsymbol{\psi} \right) = \nabla \times \boldsymbol{f}. \quad (3.57)$$

Similarly, the Helmholtz decomposition of $\boldsymbol{f}$ is given by

$$\boldsymbol{f} = \nabla \Phi + \nabla \times \boldsymbol{\Psi} \quad (3.58)$$

Thus, to satisfy the equation of motions,

$$\begin{cases} \rho \frac{\partial^2 \phi}{\partial t^2} - (\lambda + 2\mu) \nabla^2 \phi = \Phi \\ \rho \frac{\partial^2 \boldsymbol{\psi}}{\partial t^2} - \mu \nabla^2 \boldsymbol{\psi} = \boldsymbol{\Psi} \end{cases} \quad (3.59)$$

is sufficient ^{note 6)}. These equations correspond to wave equations of P-wave and S-wave, respectively.

^{note 6)} There are also terms for translational motion and rigid body rotation (see Problem 3.7) as solutions, but they can be dropped because they violate the infinitesimal assumption.

Problem 3.7

Let us consider the equation:

$$\nabla^2 \left(\frac{1}{\alpha^2} \frac{\partial^2 \phi}{\partial t^2} - \nabla^2 \phi \right) = 0 \quad (3.60)$$

1. Estimate ϕ_0 , which satisfies $\nabla^2 \phi_0(\mathbf{x}, t) = 0$.
2. Show that $\phi + \phi_0$ also satisfies equation 3.60.
3. Show that ϕ_0 represents a translational motion when ϕ_0 satisfies equation 3.53.
4. Let us consider the same discussion for the vector potential ψ_0 given by

$$\psi_0 = \begin{pmatrix} 0 & r_{xy} & r_{xz} \\ -r_{xy} & 0 & r_{yz} \\ -r_{xz} & -r_{yz} & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \quad (3.61)$$

which represents a rigid rotation. Based on the discussion for scalar potential, the vector potential satisfies the governing equation. In this text, we drop the effects because they are not first-order variables.

3.4.2 Green's function for an explosive source

In general, an excitation problem in an elastic medium by a force is complicated. In order to understand the important concept of "near field" and "far field", let us consider an excitation problem of elastic waves by an explosion source, which can be described by only a scalar potential. This simple example could be helpful for understanding elastic wave propagation.

First, let us consider Green's function G^ϕ for scalar potential ϕ . For an impulsive pressure $\delta(t)$, the Green's function is given by

$$\frac{\partial^2 G^\phi(\mathbf{x}, t; \boldsymbol{\xi}, \tau)}{\partial t^2} - \frac{1}{\alpha^2} \nabla^2 G^\phi = \delta(\mathbf{x} - \boldsymbol{\xi}) \delta(t - \tau) \quad (3.62)$$

$$G^\phi(\mathbf{x}, t; \boldsymbol{\xi}, \tau) = \frac{1}{4\pi\rho\alpha^2 |\mathbf{x} - \boldsymbol{\xi}|} \delta\left(t - \tau - \frac{|\mathbf{x} - \boldsymbol{\xi}|}{\alpha}\right) \quad (3.63)$$

Near field term and far-field term for a point explosive source

First, let us consider a response for a point explosive source for simplicity. Within an infinitesimal sphere with radius Δr at the origin suddenly increase the pressure Δp at $t = 0$ as,

$$p^{\text{source}}(\boldsymbol{\xi}, \tau) = \Delta p (1 - H(r_\xi - \Delta r)) H(\tau), \quad (3.64)$$

where r_ξ is the distance between the origin and $\boldsymbol{\xi}$, and $H(t)$ is Heaviside function^{note 7)}. The minus sign originated from the difference between pressure and stress tensor. Because an explosive source cannot cause shear deformation in a homogeneous medium, we consider only scalar potential ϕ here. ϕ can be given by convolution between the Green's function and the source term p^{source} as

$$\begin{aligned} \phi &= \int_{-\infty}^{\infty} G^\phi(\mathbf{x}, t, \boldsymbol{\xi}, \tau) p^{\text{source}}(\boldsymbol{\xi}, \tau) dV(\boldsymbol{\xi}) d\tau \\ &= -\frac{\Delta p \Delta r^3}{3\rho\alpha^2} \frac{H\left(t - \frac{r}{\alpha}\right)}{r}. \end{aligned} \quad (3.65)$$

The displacement s can be represented by gradient of the scalar potential ϕ as

$$s_r = \frac{\Delta p \Delta r^3}{3\rho\alpha^2} \left(\underbrace{\frac{H(t - r/\alpha)}{r^2}}_{\text{Near field term}} + \underbrace{\frac{\delta(t - r/\alpha)}{\alpha r}}_{\text{far field term}} \right). \quad (3.66)$$

The frequency component of the displacement $S(\omega)$ is

$$S_r(\omega) = \frac{\Delta p \Delta r^3}{3(\lambda + 2\mu)} \frac{\omega}{\alpha^2} e^{-i\omega \frac{r}{\alpha}} \left(\underbrace{-\frac{i}{(kr)^2}}_{\text{Near field term}} + \underbrace{\frac{1}{kr}}_{\text{far field term}} \right), \quad (3.67)$$

^{note 7)} $H(t) = 1, t \geq 0, H(t) = 0, t < 0$

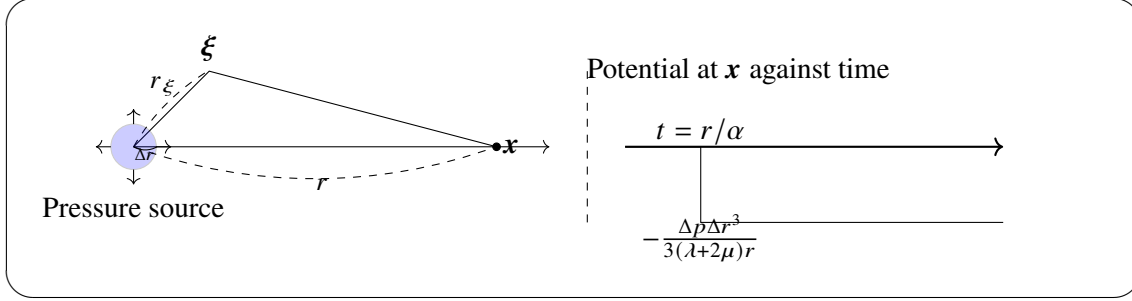


Figure 3.2: Schematic figure of the deformation by the explosive source.

This equation shows that the near-field term dominates at distances shorter than the wavelength.

The displacement of the near-field term can be interpreted as static deformation by the incremental pressure at a point. For $t = \infty$ equation 3.62 can be simplified as,

$$\nabla^2 G^\phi = -\alpha^2 \delta(\mathbf{x} - \boldsymbol{\xi}). \quad (3.68)$$

The form of this equation is equivalent to a static electric field by a point charge.

The far-field term represents a propagation of the delta function. The amplitude decreases with distance $\sim 1/r$. This result means that net energy flux on a given sphere is constant ($4\pi r^2 (r^{-1})^2$ is constant) against the propagation distance. This result originated from energy conservation owing to the wave propagation.

vspace3cm

Problem 3.8

1. Derive equation 3.65.
2. Derive equation 3.66, and plot the near field term and the far field term.

3.4.3 Green's function of a homogeneous medium for impulsive force: a general case

Green's function is feasible for estimating an elastic response by a general forcing. Let us consider Green's function for an impulsive force $X_0(t)\delta(t)$ in the x direction,

$$\frac{\partial^2}{\partial t^2} \mathbf{G} = \alpha^2 \nabla (\nabla \cdot \mathbf{G}) - \beta^2 (\nabla \times \nabla \times \mathbf{G}) + [\delta(\mathbf{x}) X_0(t), 0, 0] \quad (3.69)$$

The external force can also be represented by scalar potential Φ and vector potential Ψ based on Helmholtz theorem as,

$$[\delta(\mathbf{x}) X_0(t), 0, 0] = \nabla \Phi + \nabla \times \Psi. \quad (3.70)$$

div of both sides of the equation leads to

$$\nabla^2 \Phi = X_0(t) \frac{\partial \delta(\mathbf{x})}{\partial x}. \quad (3.71)$$

Here we consider the Green's function (equation 3.3.3) satisfies the following equation:

$$\nabla^2 \left(-\frac{1}{4\pi} \frac{1}{|\mathbf{x}|} \right) = \delta(\mathbf{x}). \quad (3.72)$$

The comparison leads to [note 8](#)),

$$\begin{aligned} \Phi &= -\frac{X_0(t)}{4\pi} \frac{\partial}{\partial x} \frac{1}{|\mathbf{x}|} \\ \Psi &= \frac{X_0(t)}{4\pi} \left(0, \frac{\partial}{\partial z} \frac{1}{|\mathbf{x}|}, -\frac{\partial}{\partial y} \frac{1}{|\mathbf{x}|} \right). \end{aligned} \quad (3.73)$$

G can be also represented by a superposition of scalar potential ϕ and vector potential ψ based on Helmholtz theorem as,

$$G = \nabla \phi + \nabla \times \psi. \quad (3.74)$$

Each potential satisfies the wave equation. For example, the scalar potential satisfies the following equation,

$$\frac{\partial^2 \phi}{\partial t^2} = \alpha^2 \nabla^2 \phi + \Phi. \quad (3.75)$$

ϕ can be given by convolution between the scalar Green's function in the 3-D medium and the forcing term Φ (see the first section of this chapter).

The scalar potential can be written by a convolution between the scalar Green's function and Φ as

$$\phi(\mathbf{x}, t) = -\frac{1}{(4\pi)^2 \rho \alpha^2} \int_V X_0 \left(t - \frac{|\mathbf{x} - \mathbf{x}_0|}{\alpha} \right) \frac{1}{|\mathbf{x} - \mathbf{x}_0|} \frac{\partial}{\partial x_0} \frac{1}{|\mathbf{x}_0|} dV_0. \quad (3.76)$$

A change of variables of $|\mathbf{x} - \boldsymbol{\xi}| = \alpha \tau$ leads to the following equation:

$$\phi(\mathbf{x}, t) = -\frac{1}{(4\pi)^2 \rho \alpha^2} \int_0^\infty \frac{X_0(t - \tau)}{\tau} \left(\iint_S \frac{\partial}{\partial \xi_0} \frac{1}{|\boldsymbol{\xi}|} dS \right) d\tau. \quad (3.77)$$

^{note 8)} see Aki and Richards⁽¹⁾ in details.

Here we focus on the integral $\iint dS$. The circle in Figure 3.3 represents a spherical shell with the radius $\alpha\tau$. The surface integral of a potential $1/r$ leads to an analogy to a problem of gravity potential. If the mass is the uniform distribution on a closed shell, the term represents the spatial gradient of the potential along x direction at the origin O (i.e., it corresponds to x component of gravity). Based on the analogy for a problem of gravity, when the point O is inside the shell, the gravity is 0. On the other hand, when it is outside the shell, the mass is concentrated at the center x virtually. This result leads to the following form:

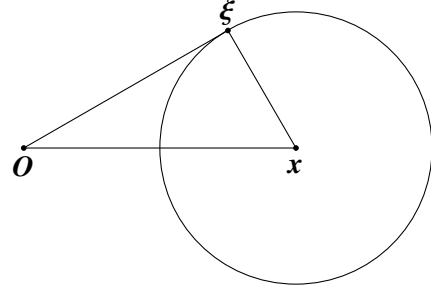


Figure 3.3: Coordinate for evaluation the potential ϕ .

$$\phi(\mathbf{x}, t) = -\frac{1}{4\pi\rho} \left(\frac{\partial}{\partial x} \frac{1}{r} \right) \int_0^{r/\alpha} \tau X_0(t - \tau) d\tau. \quad (3.78)$$

The vector potential is also written similarly. With some calculations, we obtain the potentials as,

$$\begin{cases} \phi(\mathbf{x}, t) = -\frac{1}{4\pi\rho} \left(\frac{\partial}{\partial x} \frac{1}{r} \right) \int_0^{r/\alpha} \tau X_0(t - \tau) d\tau \\ \psi(\mathbf{x}, t) = \frac{1}{4\pi\rho} \left(0, \frac{\partial}{\partial z} \frac{1}{r}, -\frac{\partial}{\partial y} \frac{1}{r} \right) \int_0^{r/\beta} \tau X_0(t - \tau) d\tau. \end{cases} \quad (3.79)$$

By substituting the scalar and vector potentials into $\mathbf{s} = \nabla\phi + \nabla \times \boldsymbol{\psi}$, we obtain the following expression:

$$\begin{aligned} s_i(\mathbf{x}, t) &= \frac{1}{4\pi\rho} \left(\frac{\partial}{\partial x_i \partial x} \frac{1}{r} \right) \int_{r/\alpha}^{r/\beta} \tau X_0(t - \tau) d\tau \\ &+ \frac{1}{4\pi\rho\alpha^2 r} \left(\frac{\partial r}{\partial x_i} \frac{\partial r}{\partial x} \right) X_0 \left(t - \frac{r}{\alpha} \right) \\ &+ \frac{1}{4\pi\rho\beta^2 r} \left(\delta_{i1} - \frac{\partial r}{\partial x_i} \frac{\partial r}{\partial x} \right) X_0 \left(t - \frac{r}{\beta} \right). \end{aligned} \quad (3.80)$$

The first term represents a near field, the second represents a far field of the P wave, and the third represents a far field of the S wave.

Now let us consider the near and far field terms in more detail. We can determine whether we are near or far based on two timescales: the characteristic time of $X(t)$, and the P-S travel time difference $r/\beta - r/\alpha$. From a simple calculation (see Problem 3.9), when the characteristic time of $X(t)$ is sufficiently smaller than the P-S travel time difference, the second and third terms are proportional to $1/r$, and the first term is proportional to r^{-2} . The first term is negligible at an enough distant point, whereas it represents the static displacement corresponding to the crustal movement at a near-source point. On the other hand, if $X(t)$ is sufficiently longer than the P-S travel time difference, all terms are proportional to $1/r$, and all terms become important. The details will be explained in the next chapter. Still, the actual earthquake is a bit more complicated,

and there are intermediate terms in addition to the far and near terms, which can be understood using Green's function derived in this section.

Problem 3.9

1. In a case of $X(t) = \delta(t)$, evaluate the near field term (the first term of the right-hand side of equation 3.80).
2. In a case of $X(t) = H(t)$, evaluate the near field term (the first term of the right-hand side of equation 3.80). Here H is the Heaviside step function.

Problem 3.10

1. When r is large enough, the far-field term of Green's function proportional to $1/r$ (the second term of equation 3.80) becomes dominant at the distant point. By evaluating the P-wave potential $\nabla\phi$ (equation 3.79), estimate the far field term of P-wave displacement.
2. Derive equation 3.80.
3. For $X(t) = \delta(t)$, evaluate the near field term (the first term of equation 3.80).

§3.5 Huygens principle

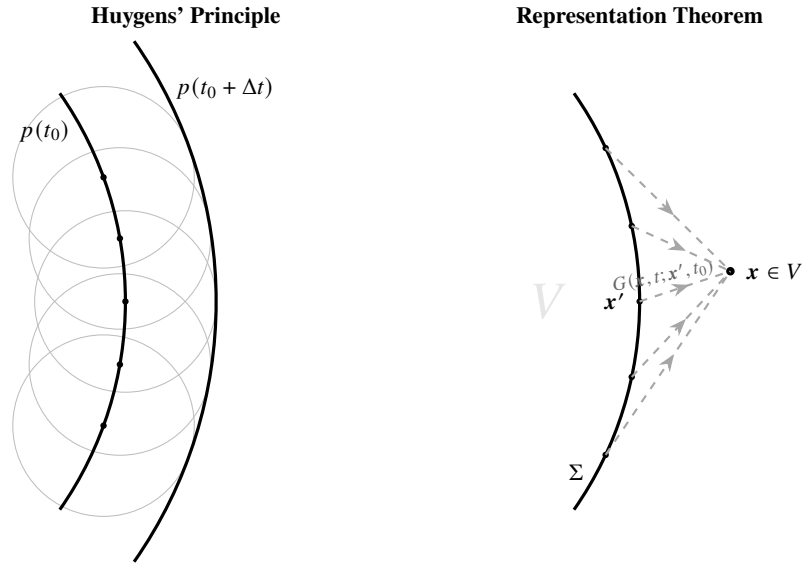


Figure 3.4: Caption

§3.6 Reciprocity of acoustic wave

Reciprocity of acoustic wave states that an acoustic wave at point A excited by a source at B is the same as an acoustic wave at point B excited by a source at A. *"If I can hear you, you can hear me."*⁽⁵⁾ The theorem is valid under a certain condition. For example, the **wind** effect breaks the theorem. In this section, I try to explain the physical and mathematical background of the reciprocity theorem of the acoustic wave.

Here we consider that an external force $F(\mathbf{x}, \omega)$ exerts on the system, and causes a pressure perturbation $P(\mathbf{x}, \omega)$.

$$\nabla \cdot \frac{\nabla P(\mathbf{x}, \omega)}{\rho_0(\mathbf{x})} + \frac{\omega^2}{\kappa(\mathbf{x})} P(\mathbf{x}, \omega) = F(\mathbf{x}, \omega) \equiv \mathcal{F} \left[\nabla \cdot \left(\frac{\mathbf{f}}{\rho_0} \right) \right]. \quad (3.81)$$

Formally, we consider a quantity PF ^{note 9)}.

^{note 9)} The multiplication of Fourier components $F(\omega)P(\omega)$ corresponds to the convolution integral of P and F in the time

Here we consider a pair of acoustic wave fields $(P_1, F_1$ and $P_2, F_2)$ ^{note 10}.

$$\nabla \cdot \frac{\nabla P_1(\mathbf{x}, \omega)}{\rho_0(\mathbf{x})} + \frac{\omega^2}{\kappa(\mathbf{x})} P_1(\mathbf{x}, \omega) = F_1(\mathbf{x}, \omega), \quad (3.82)$$

$$\nabla \cdot \frac{\nabla P_2(\mathbf{x}, \omega)}{\rho_0(\mathbf{x})} + \frac{\omega^2}{\kappa(\mathbf{x})} P_2(\mathbf{x}, \omega) = F_2(\mathbf{x}, \omega). \quad (3.83)$$

Here we consider the difference between two quantities (i) $F_1 P_2$ and (ii) $F_2 P_1$. Multiple equation 3.82 with P_2 , multiply equation 3.83 with P_1 , and subtract the resulting expressions (we call this quantity the interaction quantity ⁽³⁾). Integration over volume within Σ leads to ^{note 11}

$$\int_V (P_2 F_1 - P_1 F_2) dV = \int_{\Sigma} \frac{1}{\rho} (P_2 \nabla P_1 - P_1 \nabla P_2) \cdot \hat{\mathbf{n}} d\Sigma. \quad (3.84)$$

Here we consider a problem under a homogeneous boundary condition (on the boundary Σ , $P = 0$ or $\nabla P = 0$). The left-hand side of equation 3.84 disappears. Green's function for $F_1 = \delta(\mathbf{x})$, $F_2 = \delta(\xi)$ exhibits the spatial symmetry as,

$$G(\mathbf{x}, \mathbf{x}_1, \omega) = G(\mathbf{x}_1, \mathbf{x}, \omega). \quad (3.85)$$

This equation is known as reciprocity, which is crucial for both theoretical considerations and numerical applications.

For Earth, the ground can be approximated as a free surface ($p = 0$). I note that the Coriolis force, owing to Earth's rotation, breaks the reciprocity at very low frequency ($< 10^{-3}$ Hz). When we consider an advection term of the mean flow, such as infrasound propagation within a layer of a westerly jet, the reciprocity is also broken.

3.6.1 [†] Hermitian property of an operator

Because the mathematical operations are a bit abstract, you may find it difficult to understand the physical meanings. Here, we will try to explore the physical implications a little further. However, you may skip the following, as it anticipates the contents of the later further chapter (chapter 8).

First, the governing equations are abstracted as linear partial differential operators.

$$\mathcal{L} \equiv \nabla \cdot \frac{\nabla}{\rho_0(\mathbf{x})}. \quad (3.86)$$

Here we consider arbitrary functions u and v .

domain. Physically, this represents the time integral of a **work-related quantity** that the wave source and the wavefield mutually exert on each other in a time-reversed state.

^{note 10}) In this case, because we neglect the advection term due to the smaller amplitude, the wavefield can be characterized by pressure perturbation.

^{note 11}) de Hoop [1988]⁽³⁾ wrote "As far as acoustic wave fields are concerned, Lord Rayleigh is commonly credited as the first to derive a reciprocity theorem; it applies to harmonic sound vibrations in a homogeneous, ideal fluid. (He denotes it as Helmholtz's theorem but gives no reference to Helmholtz.)"

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The operator $\bar{\mathcal{L}}$ that satisfies

$$\left(\int u^* \mathcal{L} v dV \right)^* = \int u^* \bar{\mathcal{L}} v dV. \quad (3.87)$$

is called the Hermitian conjugate operator with $\mathcal{L}$. If $\mathcal{L}$ satisfies

$$\mathcal{L} = \bar{\mathcal{L}}, \quad (3.88)$$

we call $\mathcal{L}$ the Hermite operator or self adjoint operator ^{note 12)}.

The Hermitian operator has two important characteristics ^{note 13)}: (i) the eigenvalues are real, and (ii) the eigenfunctions are orthogonal and form a complete system. Therefore any function can be expanded in terms of eigenfunctions.

Here we consider eigenvalues λ_n and eigenfunctions u_n for an operator $\mathcal{L}$ as:

$$\mathcal{L} u_n = -\frac{\lambda_n}{\kappa(\mathbf{x})} u_n. \quad (3.89)$$

Expanding the Green's function in terms of eigenfunctions yields

$$G = \sum_n a_n u_n. \quad (3.90)$$

The insertion of the expansion into the wave equation leads to

$$\sum \left(\frac{\omega^2 - \lambda_n}{\kappa(\mathbf{x})} \right) a_n u_n = \delta(\mathbf{x} - \mathbf{x}_1). \quad (3.91)$$

From the orthonormality of the eigenfunctions:

$$\int \frac{u_n^*(\mathbf{x}) u_{n'}(\mathbf{x})}{\kappa(\mathbf{x})} dV = \delta_{nn'}, \quad (3.92)$$

the coefficients a_n are given by

$$a_n = \frac{u_n(\mathbf{x}_1)}{\omega^2 - \lambda_n}. \quad (3.93)$$

Accordingly, Green's function can be written by

$$G = \sum \frac{u_n(\mathbf{x}_1) u_n(\mathbf{x}_2)}{\omega^2 - \lambda_n}. \quad (3.94)$$

From this expression, Green's function satisfies the reciprocity property. This discussion also shows that, generally, when the eigenfunctions can be described as self-adjoint operators, the

^{note 12)}In the case of one-dimensional problems, the equations can be reduced to ordinary differential equations of the Sturm-Liouville type. In the case of horizontally stratified structures, the equation can be rewritten in the Sturm-Liouville form by separating variables. Therefore, the various properties have been studied in detail and are easy to treat.

^{note 13)}Remember the Hermitian matrix

eigenfunctions satisfy orthogonality and completeness, and therefore, the reciprocity is satisfied. For example, in the case of sound waves with mean wind, when the sound wave propagates upwind or downwind, the travel time changes due to the effect of the mean wind, and thus, the reciprocity is not satisfied. In this case, the governing equations do not satisfy the self-associative property.

Also, the definition of the Hermite operator shows that checking the interaction quantity is checking whether the self-adjoint property is satisfied. When rewritten in matrix form, the Hermitian operator corresponds exactly to the Hermitian matrix, and the various properties become easier to understand [note 14](#)).

Problem 3.11

1. Using Green's theorem, Derive equation 3.84. Hint : Use $\nabla \cdot (F\mathbf{u}) = F(\nabla \cdot \mathbf{u}) + \mathbf{u} \cdot \nabla F$, and Gauss's divergence theorem.

note 14) For example, the Hermitian matrix corresponds to an orthogonal matrix that is diagonalizable and whose eigenvalues are real numbers

3.6.2 [†]Physical interpretation of interaction quantity

Evaluating the quantity of interaction, we can derive the reciprocity. However, it is not quite clear what it means in the physical sense. Let us consider the interaction quantity, focusing on how the order of deformation changes the work done by external forces.

Total amount of work done by external forces on the fluid

First, we evaluate the work done by external forces on the fluid per unit time. The work per unit time, $w(t)$, done by external forces in the region V is given by

$$w(t) = \int_V \mathbf{f} \cdot \mathbf{v} dV \quad (3.95)$$

where $\mathbf{v}$ is the particle velocity. In this chapter, we use pressure p as the variable, so we consider the expression in terms of p instead of $\mathbf{v}$. From the equation of motion,

$$\rho_0 \partial_t \mathbf{v}(\mathbf{x}, t) = -\nabla p(\mathbf{x}, t) \quad (3.96)$$

If we define $h(t)$ as the function obtained by integrating p once with respect to time, $\mathbf{v}$ can be written as

$$\mathbf{v}(\mathbf{x}, t) = -\frac{1}{\rho_0} \nabla h(t) \quad (3.97)$$

Using the relation $\nabla \cdot (F\mathbf{u}) = F(\nabla \cdot \mathbf{u}) + \mathbf{u} \cdot \nabla F$,

$$w(t) = \int_V \left(\nabla \cdot \left(\frac{\mathbf{f}}{\rho_0} \right) \right) h dV \quad (3.98)$$

We now consider the integral of $w(t)$ over all time. From the generalized Parseval's theorem^{note 15)} we find that

$$\int_{-\infty}^{\infty} w(t) dt = \frac{1}{2\pi} \int_V \int_{-\infty}^{\infty} F^*(\mathbf{x}, \omega) \frac{P(\mathbf{x}, \omega)}{i\omega} d\omega dV$$

Here, F is the Fourier component of $\nabla \cdot \left(\frac{\mathbf{f}}{\rho_0} \right)$, and $P(\omega)$ is the Fourier component of p . In summary, by integrating the Fourier component $F^*P/(i\omega)$ over the frequency domain, we can estimate the total work done by external forces on the fluid.

^{note 15)}The generalized Parseval's theorem states that for any functions f and g ,

$$\int_{-\infty}^{\infty} f^*(t)g(t)dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} F^*(\omega)G(\omega)d\omega \quad (3.99)$$

The left-hand side can be interpreted as the cross-correlation function at lag 0, and this equation can be derived by considering the inverse transform of the cross spectrum.

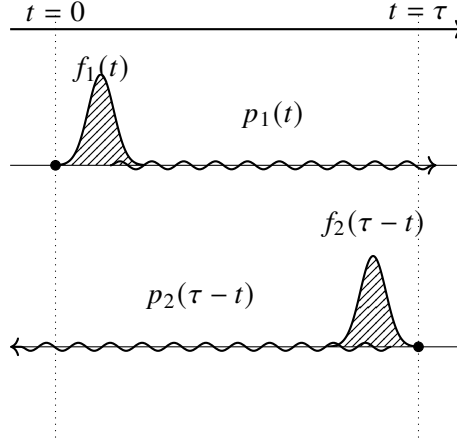


Figure 3.5: At $t = 0$, a force $f_1(\mathbf{x}, t)$ is applied, generating a sound field $p_1(\mathbf{x}, t)$. At $t = \tau$, a time-reversed force $f_2(\mathbf{x}, \tau - t)$ is applied, and we consider the sound field $p_2(\mathbf{x}, \tau - t)$ in the reverse time direction. We evaluate the total work done by the external forces. The order of applying the forces is then reversed, and the respective works are compared.

Why can we compare components at each frequency?

To compare work, we need to consider the time integral of $f(t)p(t)$ or the equivalent frequency integral of $F(\omega)P(\omega)$. However, the interaction quantity is evaluated at each frequency. Why is this? This is because there is a condition that the order in which forces f_1 and f_2 are applied does not matter.

Let us consider a situation like in Figure 3.5. Suppose a force $f_1(\mathbf{x}, t)$ is applied at $t = 0$, generating a sound wave $p_1(\mathbf{x}, t)$. At time $t = \tau$, a force $f_2(\mathbf{x}, \tau - t)$ is applied, and we consider the time-reversed sound field $p_2(\mathbf{x}, \tau - t)$. In this case, the work done by f_1 on p_2 can be written as

$$\int_V \int_0^\tau f_1(\mathbf{x}, t) p_2(\mathbf{x}, \tau - t) dV(\mathbf{x}) dt, \quad (3.100)$$

Since this equation holds for any time shift τ , the work is a function of τ . Considering the Fourier transform with respect to τ , we find that

$$\int_V F_1(\mathbf{x}, \omega) P_2(\mathbf{x}, \omega) dV(\mathbf{x}), \quad (3.101)$$

Thus, it can be decomposed into components at each frequency. Since it must hold for any time shift, we need to consider the balance of each Fourier component.

Changing the order of applying forces

- (i) Applying f_1 first to generate a sound wave, and then applying the time-reversed f_2 after τ seconds.
- (ii) Applying f_2 first to generate a sound wave, and then applying the time-reversed f_1

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after τ seconds. To compare these cases, we derive the condition $\text{Im}\{F_1 P_2 - F_2 P_1\} = 0$, focusing only on the imaginary part of the interaction quantity. Here, causality becomes important.

When considering signals that satisfy causality, causal signals must satisfy the Kramers–Kronig relations. Considering the Fourier components of causal signals, the real and imaginary parts must satisfy the Hilbert transform relations. Therefore, satisfying $\text{Im}\{F_1 P_2 - F_2 P_1\} = 0$ simultaneously satisfies $\text{Re}\{F_1 P_2 - F_2 P_1\} = 0$ ^{note 16)}, and as a result, we find that the condition for the interaction quantity to be zero ($F_1 P_2 - F_2 P_1 = 0$) must be considered.

In the chapter on Hooke’s law, we considered the symmetry of elastic constants, specifically the symmetry that does not depend on the path of deformation (this symmetry itself is known as reciprocity). When this symmetry is generalized to dynamic problems, it is shown that the reciprocity of the Green’s function for sound waves can be derived.

^{note 16)}Strictly speaking, the Fourier components of causal signals must also satisfy finiteness.

§3.7 Representation theorem: as a natural extension of Huygens's principle

The representation theorem is key for interpreting seismic wave propagation. In this section, I show the theorem is a natural extension of Huygens's principle, which does not predict the wavefront but also the amplitudes. For simplicity, we start a case of an acoustic (scalar) wave equation.

With an assumption that F_1 is 0 within a volume V ($P_1 = P$) and $F_2 = \delta(\mathbf{x} - \mathbf{x}_1)$, we obtain the equation,

$$P(\mathbf{x}_1, \omega) = \int_{\Sigma} \frac{1}{\rho} (P(\mathbf{x}, \omega) \nabla G(\mathbf{x}, \mathbf{x}_1, \omega) - G(\mathbf{x}, \mathbf{x}_1, \omega) \nabla P(\mathbf{x}, \omega)) \cdot \hat{\mathbf{n}} d\Sigma. \quad (3.102)$$

For a better description of the boundary condition, we rewrite pressure gradient by displacement $\mathbf{S}(\mathbf{x}, \omega)$ as,

$$P(\mathbf{x}_1, \omega) = \int_{\Sigma} \frac{1}{\rho} \{P(\mathbf{x}, \omega) \nabla G(\mathbf{x}, \mathbf{x}_1, \omega) - \rho \omega^2 G(\mathbf{x}, \mathbf{x}_1, \omega) \mathbf{S}(\mathbf{x}, \omega)\} \cdot \hat{\mathbf{n}} d\Sigma. \quad (3.103)$$

For a given region surrounded by a boundary Σ , the displacement and the pressure distribution on the surface Σ give us a complete description of the acoustic wavefield within the surface.

To compare Huygens's principle we replace a source at $\mathbf{x}_1$ by receiver at $\mathbf{x}$ using the reciprocity as,

$$P(\mathbf{x}_1, \omega) = \int_{\Sigma} \frac{1}{\rho} \{P(\mathbf{x}, \omega) \nabla G(\mathbf{x}_1, \mathbf{x}, \omega) - \rho \omega^2 G(\mathbf{x}_1, \mathbf{x}, \omega) \mathbf{S}(\mathbf{x}, \omega)\} \cdot \hat{\mathbf{n}} d\Sigma. \quad (3.104)$$

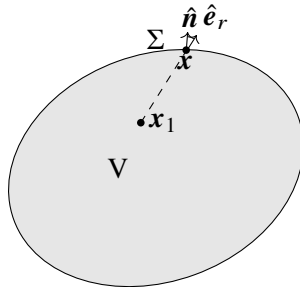


Figure 3.6: Geometry of the fluid body.

Problem 3.12

To understand the nature of the representation theorem, let us consider a one-dimensional problem for simplicity. As described in the chapter on the Green function, we consider a plate-like region $0 \leq x \leq L$ as the domain of Σ , corresponding to the case where P is the function of only x .

1. Derive the following form of the representation theorem:

$$P(x_1, \omega) = \left[\frac{1}{\rho(x)} \left\{ P(x, \omega) \left(\frac{\partial}{\partial x} G^{1D}(x_1, x, \omega) \right) - G^{1D}(x_1, x, \omega) \frac{\partial P(x, \omega)}{\partial x} \right\} \right]_{x=0}^L \quad (3.105)$$

2. In a case of a homogeneous medium ($\rho(x) = \rho_0, k = \omega/\alpha_0$), the 1-D Green's function is given by

$$G^{1D}(x_1, x, \omega) = \frac{\rho_0 i}{2k} e^{-ik|x_1 - x|}. \quad (3.106)$$

Here we consider a propagation wave solution $P(x, \omega) = P_0(\omega)e^{-ik|x|}$. Show that the representation theorem holds in this case.

The representation theorem shows that the wave field inside the boundary can be completely reproduced by information on the boundaries describing the wave inward and outward.

For a better understanding of the equation, let us consider a simple situation the Green function $G(r)$ is a function of only r using a spherical coordinate at the origin $\mathbf{x}_1$. Because $\nabla G = \partial_r G \hat{\mathbf{e}}_r$, the equation can be rewritten by,

$$P(\mathbf{x}_1, \omega) = \int_{\Sigma} \left\{ \frac{1}{\rho} P \frac{\partial G(r)}{\partial r} (\hat{\mathbf{e}}_r \cdot \hat{\mathbf{n}}) - \omega^2 G(r) (\mathbf{S} \cdot \hat{\mathbf{n}}) \right\} d\Sigma, \quad (3.107)$$

The first term of the right-hand side corresponds to a dipole source, whereas the second one corresponds to a monopole. The combination of the radiation patterns gives us a prediction of amplitudes of refracted and reflected wave fields. This formulation is known as the Fresnel-Kirchhoff diffraction formula in optics.

If you interpret the representation theorem as an extension of Huygens' principle, you have understood the physical implications. However, a closer look at the formula raises many questions. For example, if the boundary conditions do not satisfy the homogeneous boundary condition in a supposed medium (e.g., if stress and displacement are given at the boundary, as in a speaker), does the condition break the reciprocity of Green's function? This is a typical example. Actually, this problem can be avoided. That is, the wavefield combination we are considering $((F_1, P_1)$ and $(F_2, P_2))$: they are same within the region V , but not necessarily outside it.

First, as a simple case, let us consider a spherical region as the region V . Suppose that at the sphere surface, we have a free surface (satisfying the homogeneous boundary condition). In this field, you would think of a Green's function that naturally satisfies the homogeneous boundary condition on the surface of V . However, it is not necessary to choose so, and there is no problem choosing a Green's function in full space (see section 3.3.3), considering the situation where the medium continues infinitely outside the domain as well.

Next, consider the case where the homogeneous boundary condition is broken on the sphere (for example, there is a source on the boundary, such as a speaker in part). In this case, we can still choose an infinite Green's function (also called the fundamental or principal solution), and the reciprocity can be applied without any problem. In other words, the representation theorem can be applied similarly.

3.7.1 Relation to Huygens' principle

First, let us consider the boundary on a plane parallel to the wavefront in a situation where a plane wave is propagating upwards (Figure 3.7). Similar to Huygens' principle, this can be interpreted as a situation where secondary wave sources are placed on the boundary surface. The difference is that the radiation from the secondary wave sources is not uniform, but rather directed upwards. This is due to the combination of dipoles and monopoles, resulting in the radiation pattern shown in the figure.

Next, let us consider the case where the wavefront of the propagating plane wave is perpendicular to the boundary (Figure 3.8). Although it is less intuitive compared to the parallel case, let us examine it step by step. The radiation pattern of each secondary wave source is identified as a dipole. The timing of wave excitation coincides with the passage of the wavefront. Therefore, the wavefronts emitted from all secondary wave sources overlap, forming a plane wave. Since

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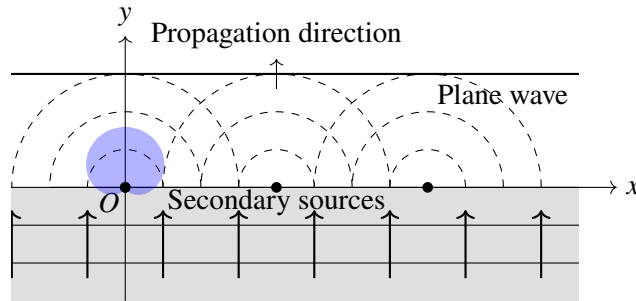


Figure 3.7: Considering the boundary parallel to the wavefront of a plane wave.

the propagation speed of the wave in the medium matches the timing of wave emission from the secondary sources, it can also be interpreted that the shock front is advancing perpendicular to the boundary.

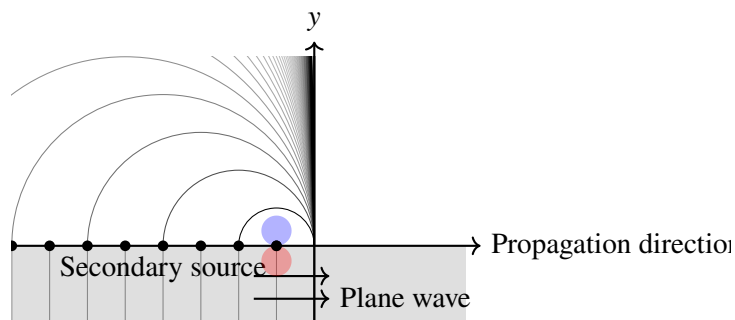


Figure 3.8: A case in which the wavefront of a plane wave is perpendicular to the boundary.

We can reconstruct all the seismic wavefields within the Earth's interior from the surface observations if we know Green's function of the medium. Based on the reconstructed seismic wavefield within the Earth, we can infer elastic constants at a given point from the ratio between the spatial gradient of pressure and the time derivative at the point. This is the basic principle of seismic imaging in seismic exploration. However, we must know the seismic structure in advance to calculate Green's function. At a glance, the logic seems to be a circular argument. From a practical point of view, we start **an initial model**, then we update it with a modification based on the surface observation and the initial Green's function. Updating the information based on the observed wavefield is essential for exploring Earth's interior using seismic exploration techniques.

Problem 3.13

Let us consider the amplitude of the plane wave excited at the boundary surface in Figure 3.7.

1. Let us consider the conditions under which the wavefront of a cylindrical wave is emphasized in Figure 3.7. For a spherical wave with radius r excited at $x = 0$, approximate the wavefront near $x = 0$ to second-order accuracy.
2. Consider the interference with the wave excited by a secondary source located at $x = \delta x$. When δx is sufficiently small compared to r , find δx such that the phase difference between the two spherical waves at $x = 0$ is approximately $\pi/4$.
3. Since we are considering the 3D Green's function, the amplitude on the wavefront is proportional to $(kr)^{-1}$. Estimate the amplitude of the secondarily excited wavefront by considering the contributions of secondary sources within the circle with a radius of δx .

Problem 3.14

Let us consider the amplitude of the plane wave excited at the boundary surface in Figure 3.8 in the same manner as Problem 3.13. Note that (i) the secondary excitation source is moving with the wavefront and (ii) the radiation pattern must be considered because the excitation sources are dipoles.

§3.8 Reciprocity of elastic medium

The reciprocity of an acoustic medium can be extended to an elastic medium. First, we extend equation 3.84 to that for an elastic medium, known as Betti's theorem. Here, I show only the results; ^{note 17)} as,

3.8.1 Betti's theorem

As in the case of the acoustic wave, we consider two types of wave fields as follows. Let us consider a pair of elastic wavefields: (i) elastic wavefield s_1 excited by forcing f_1 and (ii) elastic wavefield s_2 excited by forcing f_2 . Then integrate the inner product between the equation of motions for (i) and s' for (ii), and vice versa.

$$\int_V \{s_1(\mathbf{x}, t) \cdot (f_2(\mathbf{x}, \tau) - \rho \partial_t^2 s_2) - s_2(\mathbf{x}, \tau) \cdot (f_1(\mathbf{x}, t) - \rho \partial_t^2 s_1)\} dV \quad (3.108)$$

$$= \int_\Sigma \{[s_2(\mathbf{x}, \tau) \cdot \mathbf{T}_1(s_1(\mathbf{x}, t)) \cdot \hat{\mathbf{n}}] - [s_1(\mathbf{x}, t) \cdot \mathbf{T}_2(s_2(\mathbf{x}, \tau)) \cdot \hat{\mathbf{n}}]\} d\Sigma \quad (3.109)$$

The derivation requires the following relation,

$$\sum_{ij} E_{ij} T'_{ij} = \sum_{ij} E'_{ij} T_{ij}, \quad (3.110)$$

according to the symmetry of the elastic tensor as $C_{ijkl} = C_{klij}$ ^{note 18)}.

By time integration of the above equation with the change of variables $\tau = \tau - t$, we get the following relation:

$$\int_{-\infty}^{\infty} dt \int_V [s_1(\mathbf{x}, t) \cdot f_2(\mathbf{x}, \tau - t) - s_2(\mathbf{x}, \tau - t) \cdot f_1(\mathbf{x}, t)] dV \quad (3.111)$$

$$= \int_{-\infty}^{\infty} dt \int_\Sigma \{[s_2(\mathbf{x}, \tau - t) \cdot \mathbf{T}_1(s_1(\mathbf{x}, t)) \cdot \hat{\mathbf{n}}] - [s_1(\mathbf{x}, t) \cdot \mathbf{T}_2(s_2(\mathbf{x}, \tau - t)) \cdot \hat{\mathbf{n}}]\} d\Sigma. \quad (3.112)$$

Here we use the relation shown in Problem 3.15.

^{note 17)} read Aki and Richards⁽¹⁾ for details.

^{note 18)} This relation is based on the independence of the elastic energy on the deformation paths, as explained in the previous chapter.

Problem 3.15

Show the following relation:

$$\int_{-\infty}^{\infty} \{s_1(\mathbf{x}, t) \cdot \partial_t^2 s_2(\tau - t) - s_2(\mathbf{x}, \tau) \cdot \partial_t^2 s_1(\tau - t)\} dt = 0, \quad (3.113)$$

for given a finite time τ by partial integration under the following condition $s_1(\infty) = s_1(-\infty)$ and $s_2(\infty) = s_2(-\infty)$.

3.8.2 Reciprocity of Green's function

When we consider homogeneous boundary conditions (on a boundary Σ , $s = s' = 0$ or $T_n[s] = T_n[s'] = 0$), the right-hand side vanishes. With the transnational symmetry of Green's function, we obtain the reciprocity of the elastic Green's function as

$$G_{lm}(\mathbf{x}_2, \tau; \mathbf{x}_1, 0) = G_{ml}(\mathbf{x}_1, \tau; \mathbf{x}_2, 0) \quad (3.114)$$

$$G_{lm}(\mathbf{x}_2, \tau_2; \mathbf{x}_1, \tau_1) = G_{ml}(\mathbf{x}_1, -\tau_1; \mathbf{x}_2, -\tau_2). \quad (3.115)$$

§3.9 Representation theorem of an elastic medium

Let us insert $G_{pm}(\mathbf{x}, t; \boldsymbol{\eta}, 0)$, which represents the p component of displacement for an external force toward the m th direction at $\boldsymbol{\eta}$ in s_2 of **Betti's theorem (equation 3.8.1)**. Now, considering the translational symmetry with respect to time, the Fourier transform with respect to τ leads to the following equation

$$s_m(\omega, \mathbf{x}) = \int_V f_p(\omega, \boldsymbol{\eta}) \cdot G_{pm}(\omega, \boldsymbol{\eta}, \mathbf{x}) dV(\boldsymbol{\eta}) + \int_{\Sigma} \{G_{pm}(\omega, \boldsymbol{\eta}, \mathbf{x}) T_{pq}(s(\omega, \boldsymbol{\eta})) \hat{n}_q - s_q(\omega, \boldsymbol{\eta}) T_{pq}[(G_m(\omega, \boldsymbol{\eta}, \mathbf{x})) \hat{n}_q] d\Sigma(\boldsymbol{\eta}), \quad (3.116)$$

In these equations To obtain displacement s at $\mathbf{x}$, the above equation evaluates the convolution between Green's function for an impulsive force applied at the observed point $\mathbf{x}$ and the distributed sources. The evaluation makes the equation difficult to understand. Assuming that Green's function satisfies a homogeneous boundary condition, the reciprocity of Green's function can simplify the

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formulation ^{note 19)}. With the reciprocity, the representation theorem can be simplified as

$$s_m(\omega, \mathbf{x}) = \int_V f_p(\omega, \boldsymbol{\eta}) G_{mp}(\omega, \mathbf{x}, \boldsymbol{\eta}) dV(\boldsymbol{\eta}) + \int_{\Sigma} \{G_{mp}(\omega, \mathbf{x}, \boldsymbol{\eta}) T_p(\boldsymbol{\eta}) - s_p(\boldsymbol{\eta}) C_{pqkl} \partial_l G_{mk}(\omega, \mathbf{x}, \boldsymbol{\eta}) \hat{n}_q\} d\Sigma(\boldsymbol{\eta}), \quad (3.117)$$

where $\hat{\mathbf{n}}$ is the normal vector onto the boundary.

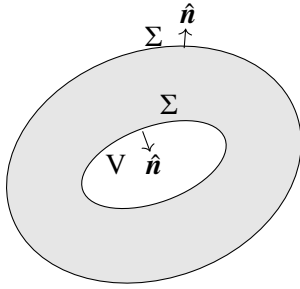


Figure 3.9: Geometry of the system we consider.

The Representation Theorem is very important when considering the excitation of seismic waves. The Representation Theorem assures us that cutting out a part of an elastic body has no effect on the motion of the elastic body outside it as long as the displacement and stress conditions at the boundary are known. Consider, for example, the phenomenon of earthquakes, which are caused by brittle fractures in a part of the Earth. The area near the fault cannot be represented by an elastic body. However, if we consider the operation of hypothetically replacing the brittle region with an elastic body through it, with a closed surface that surrounds it. Then, as long as the stresses and displacements on the boundary are identical to each other, we can completely describe the motion in the elastic body. We will see this in detail in the next chapter.

^{note 19)} The confusion arises because the boundary conditions for the Green function can be independent of those for the displacement s . This means that the choice of Green's function is arbitrary, which causes confusion. For example, when considering a homogeneous medium but complex boundaries, using the infinite medium Green's function (the fundamental solution) improves the outlook.

§3.A Delta function

3.A.1 Delta Function of a Composite Function

When $f(x)$ has the i -th zero point x_i ,

$$\delta(f(x)) = \sum_i \frac{\delta(x - x_i)}{|f'(x_i)|}, \quad (3.118)$$

This indicates that the delta function takes values at $f(x) = 0$, and it becomes clearer when considering the Taylor expansion of $f(x)$.

3.A.2 Differentiation of the Delta Function

Since the delta function cannot be differentiated in the usual sense, it is defined by integration of parts.

$$\int_{-\infty}^{\infty} f(x) \frac{d\delta(x)}{dx} dx = -f'(0) \quad (3.119)$$

3.A.3 Polar Coordinate Representation

Consider the delta function in polar coordinates r . Since $r \geq 0$, it is defined on the one hand as

$$\int_0^\epsilon \delta^+(r) dr = 1. \quad (3.120)$$

There is also a definition that sets it to $1/2$.

In three dimensions, the polar coordinate representation of the delta function is

$$\delta(\mathbf{x}) = \frac{\delta^+(r)}{4\pi r^2}. \quad (3.121)$$

Here, the delta function located at the origin is considered to be a function of r only due to its symmetry about the origin. Additionally, since the delta function can be defined through integration with a test function, integrating both sides over the entire space shows that they are equivalent.

In this case, the differentiation of the delta function is

$$\frac{\delta^+(r)}{r} = -\frac{d\delta^+(r)}{dr}. \quad (3.122)$$

This can be easily derived by integrating with the test function $g(r)/r$.

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§3.B Bessel function

$$\frac{1}{r} \frac{1}{dr} \left(r \frac{dR}{dr} \right) + \left(k^2 - \frac{m^2}{r^2} \right) R = 0, \quad (3.123)$$

is a differential equation of Bessel, and the solutions are known as Bessel function of the first kind $J_m(kr)$, and Neumann function $Y_m(kr)$. $J_0(0) = 1$, $J_m(0) = 0$, $m \neq 0$, whereas Neumann function diverges at $r = 0$. Both functions converge to $1/\sqrt{r}$ for $r \rightarrow \infty$. Read Mathematical Methods for Physicists⁽²⁾ for details.

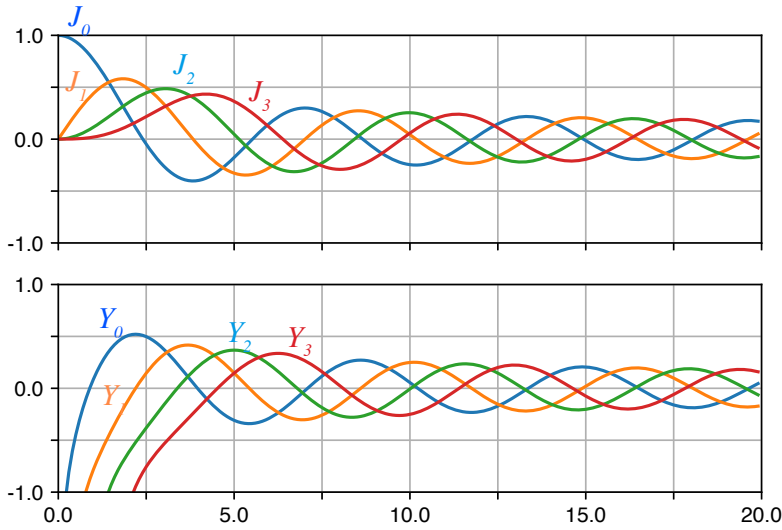


Figure 3.10: Plots of Bessel functions and Neumann functions.

3.B.1 Properties

$$J_{-m}(x) = (-1)^m J_m(x) \quad (3.124)$$

$$J_{m-1}(x) + J_{m+1}(x) = \frac{2m}{x} J_m(x) \quad (3.125)$$

$$J_{m-1}(x) - J_{m+1}(x) = 2J'_m(x) \quad (3.126)$$

3.B.2 Asymptotic for $x \rightarrow 0$

$$J_m(x) \sim \frac{1}{m!} \left(\frac{x}{2}\right)^m \quad (3.127)$$

$$Y_0(x) \sim \frac{2}{\pi} \ln \frac{x}{2}, \quad (3.128)$$

$$Y_m(x) \sim -\frac{(m-1)!}{\pi} \left(\frac{2}{x}\right)^m, \quad m \geq 0 \quad (3.129)$$

3.B.3 Asymptotic for $kr \gg 1$

$$J_m(kr) \sim \sqrt{\frac{2}{\pi kr}} \cos\left(kr - \frac{2m+1}{4}\pi\right) \quad (3.130)$$

$$Y_m(kr) \sim \sqrt{\frac{2}{\pi kr}} \sin\left(kr - \frac{2m+1}{4}\pi\right) \quad (3.131)$$

$$H_m^{(1)}(kr) \sim \sqrt{\frac{2}{\pi kr}} e^{i(kr - \frac{2m+1}{4}\pi)} \quad (3.132)$$

Here $H_m^{(2)}(kr) = H_m^{(1)*}(kr)$.

§3.C Hankel Functions

Here, we summarize the relations used in the main text regarding Hankel functions.

3.C.1 Recurrence Relations

$$\left(\frac{1}{z} \frac{d}{dz}\right)^k \left(z^{-\nu} H_\nu^{(2)}(z)\right) = (-1)^k z^{-\nu-k} H_{\nu+k}^{(2)}(z)$$

Refer to DLMF (<https://dlmf.nist.gov/10.6>).

3.C.2 Relation to 2-D Green's function

The 2-D Green's function in the frequency domain ($\omega > 0$) can be written as

$$G^{2D}(r, \omega) = -\frac{\rho_0 i}{4} H_0^{(2)}(\omega(r/c)) = -\frac{\rho_0}{4} (Y_0(\omega(r/c)) + J_0(\omega(r/c))i) \quad (3.133)$$

where $c \equiv \sqrt{\kappa/\rho_0}$. Since it must be real in the time domain, $G^{2D}(r, \omega) = (G^{2D}(r, -\omega))^*$ is required. Therefore,

$$G^{2D}(r, -\omega) = -\left(\frac{\rho_0 i}{4} H_0^{(2)}(\omega(r/c))\right)^* = -\frac{\rho_0}{4} (Y_0(\omega(r/c)) - J_0(\omega(r/c))i) \quad (3.134)$$

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The real part of the Green's function is $Y_0(|\omega|(r/c))$, which is an even function of frequency, and the imaginary part is $J_0(|\omega|(r/c)) \text{sign}(\omega)$, which is an odd function. Note that intuitively we define $J_0(x) = J_0(-x)$ for negative values. Furthermore, since the Green's function must be causal, it must satisfy the Kramers–Kronig relations, and the Hilbert transform of $J_0(\omega(r/c)) \text{sign}(\omega)$ becomes $Y_0(|\omega|(r/c))$.

3.C.3 Relation to Spherical Hankel Functions

There is a relation between the spherical Hankel function $h_n^{(2)}$ and the Hankel function $H_{\frac{n}{2}-1}^{(2)}$:

$$h_n^{(2)}(r) = \sqrt{\frac{\pi}{2r}} H_{\frac{n}{2}-1}^{(2)}$$

Additionally,

$$h_0^{(2)}(kr) = -\frac{1}{kr} e^{-ikr}$$

§3.D Plane Wave Expansion

A plane wave can be expressed as a superposition of cylindrical waves.

$$e^{ikr \cos \phi} = \sum_{n=-\infty}^{\infty} i^n J_n(kr) e^{im\phi} \quad (3.135)$$

§3.E Fourier transform

For time series $u(t)$, the Fourier transform $\mathcal{F}$ and the inverse Fourier transform $\mathcal{F}^{-1}$ are defined as

Definitions of Fourier transform

$$U(f) \equiv \mathcal{F}(u) = \int_{-\infty}^{\infty} u(t) e^{-i2\pi f t} dt, \quad (3.136)$$

$$u(t) = \mathcal{F}^{-1}(U) \equiv \int_{-\infty}^{\infty} U(f) e^{i2\pi f t} df, \quad (3.137)$$

$$(3.138)$$

where U represents Fourier components of u .

Summary of Fourier transform

- If $u(t)$ is a real function, $U(f) = U^*(-f)$,
- Parseval's theorem : $\int_{-\infty}^{\infty} u(t)^2 dt = \int_{-\infty}^{\infty} U(f)^2 df$,
- Cross spectrum $C(u, v; f) = \mathcal{F}(\psi) = U^* \tilde{v}$
- Wiener-Khinchin theorem : $p(f) = \mathcal{F}(\phi) = |U|^2$.

§3.F Hilbert transform

Hilbert transform of $f(t)$ $\mathcal{H}f(t)$ is defined by

$$\mathcal{H}f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} [-i \operatorname{sign}(\omega)] F(\omega) e^{-i\omega t} d\omega. \quad (3.139)$$

This can be interpreted as a phase advance of 90° to the original signal in the frequency domain. In the time domain, it can be written as,

$$\mathcal{H}f(t) = \frac{1}{\pi} \mathcal{P} \int_{-\infty}^{\infty} \frac{f(\tau)}{\tau - t} d\tau. \quad (3.140)$$

Here $\mathcal{P} \int$ is Cauchy's principle integral

Additionally, applying the Hilbert transform twice results in

$$\mathcal{H}[\mathcal{H}f] = -f(t), \quad (3.141)$$

which inverts the sign.

For details, read textbooks of applied mathematics (e.g., Mathematical Methods for Physicists⁽²⁾).

§3.G Kramers–Kronig Relations

Now, let us consider a signal $f(t)$. Assuming this signal satisfies causality, we consider the signal $f(t)h(t)$, where $h(t)$ is the Heaviside step function

$$h(t) = \begin{cases} 0, & t < 0, \\ 1/2, & t = 0, \\ 1, & t > 0, \end{cases} \quad (3.142)$$

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Let us consider the Fourier transform of $f(t)h(t)$. Denoting the Fourier transforms of $f(t)$ and $h(t)$ as $F(\omega)$ and $H(\omega)$ respectively, the Fourier transform of $f(t)h(t)$ can be expressed as the convolution

$$\mathcal{F}[fh] = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega') H(\omega - \omega') d\omega' \quad (3.143)$$

Here, $H(\omega)$ is given by

$$H(\omega) = \pi\delta(\omega) + \frac{1}{i\omega}. \quad (3.144)$$

The first term arises because the average value of the Heaviside function is not zero, thus it has a value at zero frequency to account for the average shift. The second term represents the integral in the frequency domain, as the Heaviside function can be roughly considered as the time integral of the delta function, resulting in $(i\omega)^{-1}$ in the frequency domain.

Substituting this into the equation, we get

$$\mathcal{F}[fh] = \frac{1}{2}F(\omega) - \frac{1}{2}i\mathcal{H}[F] \quad (3.145)$$

Denoting the real and imaginary parts of F as F_r and iF_i respectively, we have

$$\mathcal{F}fh = \frac{1}{2}(F_r(\omega) + \mathcal{H}F_i) + i\frac{1}{2}(F_i(\omega) - \mathcal{H}F_r) \quad (3.146)$$

Therefore, the imaginary part is the Hilbert transform of the real part.

In summary, causal signals have their real and imaginary parts linked by the Hilbert transform (known as the Kramers–Kronig relations). In the time domain, since the value is zero for $t < 0$, considering the degrees of freedom, there is a unique relationship between the real and imaginary parts, which is the Hilbert transform.

Problem 3.16

Let us consider the Fourier transform of the Heaviside function. Since it becomes 1 as $t \rightarrow \infty$, it cannot be Fourier transformed in the usual sense. Therefore, for sufficiently large T , we consider the function

$$h^T(t) = \begin{cases} 0, & t < 0, \\ 1/2, & t = 0, \\ 1, & 0 < t < T, \\ 1/2, & t = T, \\ 0, & t > T, \end{cases} \quad (3.147)$$

Calculate the Fourier transform of this function and then consider the limit as $T \rightarrow \infty$, showing that

$$H(\omega) = \pi\delta(\omega) + \frac{1}{i\omega} \quad (3.148)$$

§3.8 Bibliography

- [1] K. Aki and P.G. Richards. *Quantitative Seismology*. Univ Science Books, 2nd edition, 2009.
- [2] G.B. Arfken and H.J. Weber. *Mathematical Methods for Physicists*. Elsevier Science, 2013.
- [3] Adrianus T de Hoop. Time-domain reciprocity theorems for acoustic wave fields in fluids with relaxation. *J. Acoust. Soc. Am.*, 84(5):1877–1882, November 1988.
- [4] Masanori Saitō. *The theory of seismic wave propagation*. 2016.
- [5] Roel Snieder and Kasper van Wijk. *A Guided Tour of Mathematical Methods for the Physical Sciences*. Cambridge University Press, 3 edition, 2015.